Who Wins and Loses in a Bubble? Evidence from the British Bicycle Mania

William Quinn and John D. Turner

How do different types of investors perform during financial bubbles? Using a rich archival source, we explore investor performance during the British bicycle mania of the 1890s. We find that directors and employees of cycle companies reduced their holdings substantially during the crash. Those holding shares after the crash were generally not from groups stereotypically thought of as naïve, but gentlemen living near a stock exchange, who had sufficient time, money, and opportunity to engage in speculation. Our findings suggest that the investors most at risk of losing during a bubble are those prone to familiarity and overconfidence biases.

Who are the winners and losers in an asset price bubble? In the case where markets are efficient and prices follow an unbiased and unpredictable random walk, then it is unlikely that any group will significantly outperform any other (Malkiel 2003). However, this can be changed by the presence of heterogeneous information, which might allow informed or well-connected investors to "ride" the bubble (Abreu and Brunnermeier 2002, 2003; Temin and Voth 2004). Groups that tend to lose out might simply be noise traders, but it could also be vulnerable demographics, those with the least experience or information, or those with a strong preference for risky assets. Alternatively, the biggest losers could be those most vulnerable to behavioral biases, such as overconfidence or familiarity bias (Barber and Odean 2001; Seasholes and Zhu 2010).

This paper investigates this question using a new dataset of investors during and after an asset price reversal in British bicycle companies in 1895–1900. Cycle company shares experienced a substantial price

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reversal in this period, almost trebling in value in the early months of 1896 before losing 73 percent of their peak value by the end of 1898. The scale of these price movements is similar to other infamous stock market reversals: the dot-com boom saw the NASDAQ index rise 110 percent between December 1996 and its peak in March 2000, before losing 77 percent of its value by October 2002 (Quinn and Turner 2020, pp. 157–59). Like the dot-com era, the cycle mania was accompanied by a promotion boom: between January 1896 and June 1897, 601 new cycle corporations were established (Quinn 2019, p. 276).

The key advantage of studying the bicycle mania is that companies in this era were legally required to publish annually the names, occupations, addresses, and number of shares held by each shareholder. This makes our dataset complementary to other studies of shareholder clientele changes during an asset price bubble, which typically have much more frequent observations, but much less detail on shareholder identities. For example, Brunnermeier and Nagel (2004) limit their study to hedge funds, Temin and Voth (2004) investigate the holdings of one private bank, and Greenwood and Nagel (2009) study mutual fund managers with age used as a proxy for experience. Griffin et al. (2011) study a broad range of investors but can only distinguish between individuals and various types of institutional investors. For the cycle mania, we have detailed data on the occupations and addresses of all investors in each company in our sample. This allows for a much more granular observation of investor identities, especially at the less experienced end of the spectrum.

We found shareholder registers from the U.K.'s National Archives for 25 cycles, tube and tyre companies at two distinct points in time during the asset price reversal. The first time period chosen is prior to the crash, when the prices of cycle shares had not yet peaked. The second time period was during the crash, when share prices had peaked and were falling. Since all 25 of the companies in the sample were disbanded on unfavorable terms to shareholders within a decade, investors holding shares at this stage are almost certain to have lost money on their investments. There is therefore little risk of capturing investors who successfully "bargain hunted" at the nadir of a cycle. Conversely, investors who held shares prior to the crash, but were absent from the register when prices were falling are much more likely to have profited from the bubble. From each shareholder register, we record the occupation, address, and number of shares held by each investor. We also record whether the investor was a director of the company by checking their names against those listed in the *Stock Exchange Yearbook* and *Birch's Manual of Cycle Companies (1897)*.

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A large minority of cycle company shareholder registers included all share transfer information over the previous year. These registers recorded the date on which any shares were sold from one investor to another, the number of shares sold, and the name, occupation, and address of the seller (but no information on the buyer). Such registers were found for 10 of these companies, resulting in a dataset of 1,996 transfers.

In order to identify the extent to which changes in ownership can plausibly be attributed to the bubble, we also collect this data for a control group. This control group consists of 11 companies that were established between 1895 and 1898, categorized as miscellaneous by *Investor's Monthly Manual*, and had not experienced a share price crash at the time of their second surviving shareholder register. The aforementioned data on occupation, address, directorship, and transfers of shares was also collected for the control group firms. This sample provides some indication of how ownership of new companies at this time might be expected to change in the absence of an asset price reversal or crash.

Our data is first used to establish the characteristics of investors during the initial stage of the cycle boom. Relative to the control group, we find that cycle shares attracted a high level of investment from manufacturers, financiers, institutional investors, and professional middle classes, and a low level of investment from gentlemen (i.e., a social class in Britain at the time consisting of men sufficiently wealthy that they did not need an occupation) and women. This suggests that cycle investors came from groups that previous research has associated with a preference for riskier investments, but not from groups associated with a low level of investment experience (Acheson, Campbell, and Turner 2017; Rutterford et al. 2011).

The paper then examines how the shareholders of cycle firms changed over the course of the cycle mania. Cycle agents and manufacturers, and directors of the company in which they held shares, decreased their holdings during the crash, suggesting that investors with privileged information performed better during the bubble. In terms of who was left holding shares during the crash, the most striking finding is a significant increase in the number of gentlemen, particularly gentlemen based near a stock exchange. Transfer data shows that investors with these characteristics were highly active sellers, and the increase in their number implies that they were also highly active buyers.

We use a two-way difference-in-differences approach to determine which of these trends were also present in the control companies. Similar to the cycle firms, gentlemen living near a stock exchange were highly active traders in the control firms. However, the increase in the number

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of gentlemen shareholders in the years following incorporation was very small relative to the cycle firms, and the number of investors living near a stock exchange actually fell over time. A notable feature of the control group registers relative to the cycle companies was an increase in the number of female investors, who appear to have invested in these relatively successful firms as they became more established. In contrast with the cycle firms, the proportion of shares held by directors in the control firms increased over time.

These results suggest that cycle company investors with access to privileged information, such as cycle industry workers and directors of the company in which they held shares, were able to exit prior to the crash, often making a profit on their investment. The biggest losers in the bubble were men with considerable wealth, plenty of free time, and a convenient nearby stock exchange, who were often very active traders. The risk of losing money during a bubble thus appears to be linked to having sufficient time, money, and opportunity to trade shares frequently. This finding contradicts popular anecdotes about financial bubbles, which portray latecomers as members of the working class, such as shoeshine boys, busboys, or taxi drivers (Kindleberger 1978). It is, however, consistent with the demographic profile of speculative day traders in modern markets, which typically skews old, wealthy, and male (Arthur and Delfabbro 2017).

These results can be interpreted as a manifestation of familiarity bias, whereby investors experience lower risk-adjusted returns as a result of a preference for investments they are familiar with (Goetzmann and Kumar 2008). For example, investors tend to prefer the stocks of local firms, firms they are customers of, and firms they work for, possibly due to the illusion of superior information (Benartzi 2001; Coval and Moskowitz 1999; Grinblatt and Keloharju 2001; Huberman 2001; Seasholes and Zhu 2010). In this case, gentlemen living near a stock exchange may have believed that this proximity gave them an informational advantage that would allow them to beat the market, when in reality this was not the case.

This paper contributes to the growing literature examining the behavior of investors during asset price booms. Our study is complementary to previous studies on the subject, which typically have much more frequent observations of ownership, but significantly less detail on shareholder identities (Acheson, Aldous, and Quinn 2023; Brunnermeier and Nagel 2004; Frehen, Goetzmann, and Rouwenhorst 2013; Griffin et al. 2011). Our results provide new insights into the types of investors that gain or lose money during a bubble. While the most informed investors appear

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to perform best, the worst-performing investors were not those with the least information, but those with just enough information to make them overconfident in their ability to profit from the bubble.

THE BICYCLE MANIA

In the 1880s and early 1890s, there were a series of technological innovations in the production of bicycles, most notably the pneumatic tyre, weldless steel tube, and diamond frame (Harrison 1969).[1] The sharp improvement in the quality and cost of bicycles led to a rapid increase in demand, which came to a head in the "bicycle boom" of 1895–1897 (Amini and Toms 2018). It is estimated that, at its peak, 750,000 bicycles were being produced annually, with 1.5 million people cycling out of a U.K. population of around 35 million (Rubinstein 1977, p. 51). The cycle industry went into recession after 1897, with Harrison (1969) attributing its decline to American competition, the over-capitalization of many cycle firms, and bicycles going out of fashion.

The boom and bust observed in the cycle industry was mirrored by a reversal in the shares of cycle companies. Two events in April 1896 catalyzed a rapid increase in the share prices of cycle companies. First, the Pneumatic Tyre Company was purchased for £3 million, ten times its nominal capital, and successfully recapitalized for £5 million as the Dunlop Pneumatic Tyre Company. At almost exactly the same time, the Beeston Pneumatic Tyre Company announced its intention to pay a dividend of 100 percent for the year 1896. In response, trading on the Birmingham Stock Exchange, where most cycle firms listed, was said to have "gone mad."[2] Prices appear to have been sustained until mid-1897, when financial newspapers were warning investors in cycle shares of an upcoming slump.[3] Newspaper reports from 1898 suggest that the cycle share market had collapsed.[4]

Figure 1 shows the value of a cycle share index between September 1895 and December 1898, alongside the number of companies whose share price was reported. The index rose from a value of 88 in January 1896 to a peak of 250 in May of the same year, increasing by 184 percent. Share prices then declined throughout 1896, but experienced a second boom in early 1897. From mid-March 1897 onward, prices continuously declined until the end of 1898: the index fell from 241 in March 1897 to

[1] For previous accounts of the cycle mania, see Harrison (1969, 1981), Rubinstein (1977), Millward (1989), Lloyd Jones and Lewis (2000), and Quinn and Turner (2020).

[2] *Financial Times*, "The Cycle Trade Boom," 22 April 1896.

[3] *Financial Times*, "The Cycle Outlook," 1 May 1897.

[4] *Financial Times*, "The Slump in the Cycle Trade," 15 June 1898.

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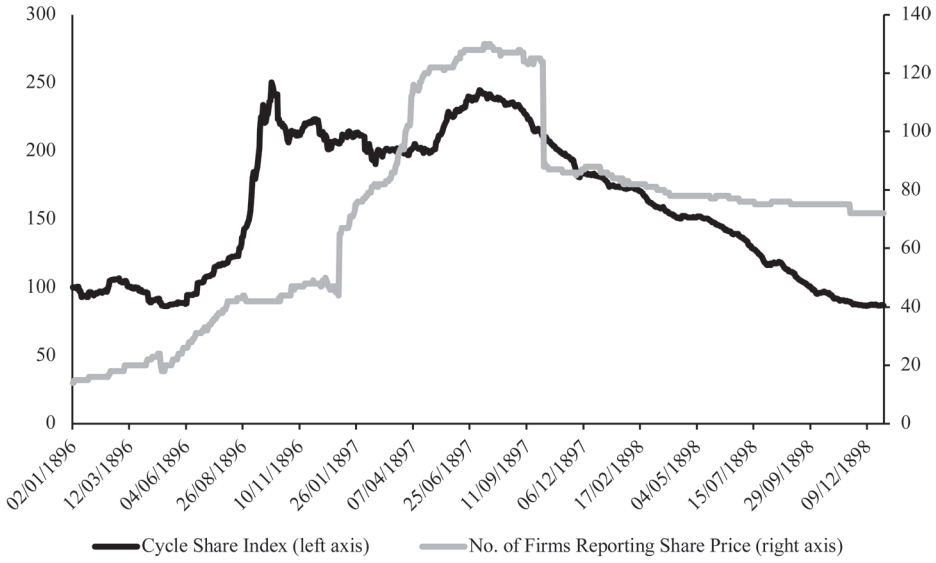


FIGURE 1
CYCLE SHARE INDEX AND NUMBER OF ACTIVELY TRADED CYCLE FIRMS

Sources: *Financial Times* and *Birmingham Daily Mail*.

66 in January 1899, losing 73 percent of its value. The number of share prices reported fell from a peak of 130 to 72 in this time, largely as a result of declining stock market activity and firms ceasing operations. Quinn (2019) investigates whether these price movements constitute an asset price bubble, finding that the scale of the price movements is not justified by earnings, future dividends, idiosyncratic risk, or plausible future expectations. In any case, the contentious question of whether prices are consistent with fundamental values during a suspected bubble is orthogonal to the hypotheses explored in this paper.

HYPOTHESES AND DATA

Data from this episode are used to examine competing hypotheses about which clientele make or lose money during a bubble. In the case of a fully efficient market, no particular group should be able to systematically outperform any other. However, recent studies have argued that those with privileged information, such as industry insiders and company directors, can sell prior to the crash, possibly even timing their sales to maximize profits (Abreu and Brunnermeier 2002, 2003). Greenwood and Nagel (2009), Kindleberger (1978), Smith et al. (1988), and Vissing-Jorgensen (2003) have argued that an influx of inexperienced or naïve

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investors can cause assets to become overvalued. If this were the case, a large proportion of shares during a mania might be held by investors from groups associated with low levels of experience and information. This might be particularly true in the later stages of the bubble, when shares had lost most or all of their value.

Another possibility is that the groups that lose out are those most vulnerable to particular behavioral biases. Prior literature has shown that many investors prefer to invest in firms they are more familiar with, and that this tends to result in lower-than-expected risk-adjusted returns (Huberman 2001). This familiarity bias can take the form of a preference for domestic stocks, a preference for geographically local stocks over other domestic stocks, or a preference for the shares of the company where the investor is employed (Cao et al. 2011; Coval and Moskowitz 1999; Grinblatt and Keloharju 2001; Døskeland and Hvide 2011). Familiarity bias has been linked with overconfidence bias: investors may believe that their proximity to the investment gives them an informational advantage, when in reality this is not the case (Døskeland and Hvide 2011). Given these findings, one might expect the biggest losers in a bubble to be investors who are close geographically or professionally to the companies involved. They might also be disproportionately drawn from demographics associated with overconfidence, such as young men (Barber and Odean 2001; Grinblatt and Keloharju 2009).

The data used to explore these possibilities comes from the *Summaries of Capital and Shares* of cycle companies, which were published annually by every publicly traded company in this era as required by the 1862 Companies Act. These documents listed the names, addresses, and occupations of shareholders in each company, alongside the number of shares held. Not every annual register for every cycle company has been preserved in the archives, so it was only possible to obtain a sample of investors before and after the peak of the bubble for 25 cycle companies, just over a quarter of the actively traded cycle firms quoted in the *Financial Times* in March 1897. Full details of these companies are listed in Appendix Table 1. These companies had an average subscribed capital of £73,418, somewhat smaller than the £159,500 for the population of listed cycle companies in Quinn (2019). However, this is largely because our sample does not include any of the largest firms, particularly the Dunlop Company, which had a subscribed capital of £4,547,000. The median actively traded cycle company had a subscribed capital of £75,000, close to the mean of our sample.

Importantly, none of the companies in the sample were long-term successes for shareholders: 11 declared bankruptcy in the aftermath of

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the crash, and the remaining 14 were reconstructed or wound up on terms that imposed heavy losses relative to par value. This was typical of cycle mania firms: of 141 actively traded firms, 113 had ceased to exist by 1910, and final share prices suggested heavy losses for shareholders (with the notable exception of firms that were acquired during the boom phase). The minority of firms that were successful in the long term still experienced substantial share price crashes during the 1900s (Quinn 2019, p. 285).

For each of the 25 companies in our sample, the occupations and addresses of shareholders are recorded at two points in time. The first point in time is either before or during the "bubble" period, ranging from November 1894 to the beginning of April 1897. Although share prices in the overall cycle share market peaked in March 1897, the two companies whose initial shareholder register is from April 1897 were still trading at a price very close to their peak value at this time. The second shareholder register of each company is taken from a point in time during the crash, ranging from the end of April 1897 to November 1900. The pre-peak and post-peak points in time, which are hereafter denoted as t_1 and t_2, respectively, provide cross-sections of investor occupations at different stages of the asset price reversal, indicating the change in shareholder clientele across time.

Figure 2 shows the dates on which the various shareholder records in our sample were published, alongside a monthly price-weighted cycle share index. Notably, the price level of cycle shares continued to fall until several months after the majority of t_2 summaries had been published. This, combined with the winding-up data in Appendix Table 1, shows that shareholders at time t_2 would almost certainly have undergone losses on their investment regardless of the precise point at which the shares were purchased. It is therefore unlikely that we are capturing some informed value investors who entered the market late.

The data for each company is aggregated, and observations where the occupation was either missing or unintelligible are removed. This results in a dataset of 5,118 and 7,049 shareholders at times t_1 and t_2, respectively, accounting for the ownership of around 1.6 million shares at t_1 and around 1.5 million at t_2. These records provide a high level of detail on investor occupations: among the 12,167 individuals in the holdings data, there are 1,241 unique occupations listed. Few investor addresses were left blank or unintelligible, but at times only a city or postcode region was provided, rather than an exact address. Investors are therefore categorized according to the postcode region or city in which they lived, rather than by distance from a significant landmark, a measure that previous studies have used (Fjesme, Galpin, and Moore 2019).

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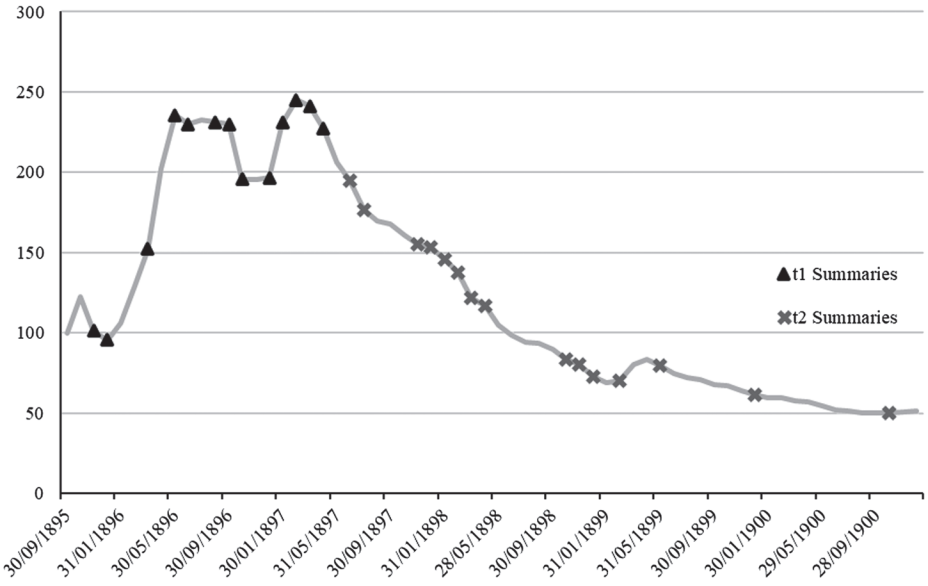


FIGURE 2
CYCLE SHARE PRICES AND SUMMARIES OF CAPITAL AND SHARES

Notes: t_1 is defined as the period of time prior to or during the peak of the company's share price, with t_2 defined as the period of time after its share price had peaked.
Sources: *Birmingham Daily Mail, Birmingham Daily Post, Financial Times,* and *Summaries of Capital and Shares.*

In order to account for potential changes in the occupations of individual shareholders, the full name of the ten largest shareholders in each company at time t_1 was also recorded. For some companies, more than ten shareholders were identified by name, because more than one investor held the tenth-most number of shares. This provided the identities of 273 investors, accounting for 83 percent of the total capitalization of these companies at t_1. These identities were then looked up in the shareholder registers at time t_2, in order to check whether their occupations had been reported differently. The listed occupations were found to be identical for 71 percent of these large investors. For the remaining 29 percent of observations, the occupation listed at time t_1 is treated as the investor's occupation at both points in time. The exception to this was when no occupation was listed at time t_1, in which case the occupation listed at time t_2 is treated as the investor's occupation throughout.

The names of the directors of each individual company were then collected from the *Stock Exchange Yearbooks* of 1896–1899. These names were looked up in the shareholder register of each company at both t_1 and t_2, in order to determine whether there was a systematic change in the number of shares held by directors. Comparing the names listed in

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TABLE 1
SUMMARY OF CYCLE COMPANY TRANSFERS

	Transfers	Shares	Average Price per Transfer (£)	Average Price per Share (£)	Maximum Price (£)	Minimum Price (£)
At or above par	978	146,979	1.29	1.27	1.86	1.00
Below par	919	367,929	0.46	0.30	0.98	0.04
Unknown price	99	43,651	—	—	—	—
Total	1,996	558,559	0.92	0.58	1.86	0.04

Sources: Share prices obtained from the *Financial Times*, transfers obtained from National Archives, *Summaries of Capital and Shares*.

various editions of the *Stock Exchange Yearbook* indicates that company directors rarely changed between t_1 and t_2, which is unsurprising given the relatively short length of time between shareholder registers. Directors typically accounted for a very low number of shareholders, but a significant proportion of shares: each company had between three and seven directors, but on average they held 25.6 percent of the company's shares at time t_1 and 20.1 percent at time t_2.

A minority of companies also recorded any sales made by shareholders since the previous summary had been published, and the dates on which these sales occurred. Records covering the relevant time period were found for 10 cycle companies, resulting in a sample of 1,996 transfers involving 558,559 shares. The dates of these transfers range from May 1896 to December 1898. For each transfer, we record the occupation of the investor selling shares, the number of shares sold, and the date on which the shares were sold.

The market price of the shares on the date of each transfer, as listed in the *Financial Times*, is also recorded and summarized in Table 1. Prices were not always quoted, and so this data was only available for 1,897 of 1,996 transfers. The summaries of capital and shares were selected to document trades that occurred while cycle share prices were relatively high, so this cannot be treated as a representative sample of all trades in cycle shares throughout the period. There is, however, notable variation in the prices at which transactions occurred: some were sold for a price 86 percent greater than the IPO subscription price, while others were sold for 96 percent less. Notably, significant numbers of investors would have been able to profit from the bubble.

A significant limitation of observing raw changes in the number of investors from a particular occupational or geographic group is that there is no baseline for comparison. This makes it impossible to determine whether the observed changes were typical of a company in the years

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following its establishment at this time, or can plausibly be attributed to the bubble. In order to address this limitation, we also gather shareholder identity data from the same source on a control group of non-cycle companies. These companies were selected based on:

1. Being classified as miscellaneous by *Investors Monthly Manual*, the same categorization as bicycle firms.
2. Being established in the 1895–98 period.
3. Having a surviving initial shareholder register and another surviving register from within the subsequent five years.
4. Not experiencing a share price bubble or crash of 50 percent or more between these two registers.

This resulted in a control sample drawn from 11 companies, which operated in industries such as tea, department stores, electricity, sulphur, hotels, and dairy products. This sample consists of 4,905 shareholders at t_1 and 5,687 at t_2, a total of 10,586. The aforementioned information on occupation, location, and number of shares held was collected for these groups, and company directors were identified from *Stock Exchange Yearbooks*.

Summary statistics of each sample of companies are shown in Table 2, and further details of each company are included in the Appendix. It can be seen that the non-cycle companies are not a perfect control group: in particular, they had a somewhat larger subscribed capital than the cycle firms. The company offices of all 11 non-cycle companies were in London, compared to only 7 of the 25 cycle companies.[5] The average time between registers is also longer for the control companies: 54 months, compared to 22 months for the cycle firms. However, the purpose of the sample is to rule out plausible alternative explanations for changes in cycle shareholder clientele so that we can reasonably attribute these changes to the bubble. For example, in the absence of a control group, our analysis may capture investors who routinely invested in IPOs and sold at the first opportunity regardless of company performance. For this purpose, the larger subscribed capital is unlikely to be an issue. The longer time between registers will, if anything, provide a better indication of how shareholder clientele might have changed in the absence of a bubble.

[5] For robustness, we repeated the subsequent analysis after omitting the two largest non-cycle firms, which reduced the average nominal capital to a similar level in both samples. We also repeated the analysis using only London-based cycle firms. In both cases, the results were materially unchanged and remained statistically significant.

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TABLE 2
COMPANY-LEVEL SUMMARY STATISTICS FOR HOLDINGS DATA

Cycle Companies				
	Average	Std. Dev.	Max.	Min.
		t_1		
Share price (% of par)	1.09	0.20	1.5	0.75
No. of shareholders	281	305	1,290	18
Subscribed capital (£)	73,418	66,871	249,837	4,291
		t_2		
Share price (% of par)	0.37	0.30	1.06	0.05
No. of shareholders	399	468	2,068	24
Subscribed capital (£)	70,003	65,330	250,000	4,500
Control Companies				
	Average	Std. Dev.	Max.	Min.
		t_1		
Share price (% of par)	1.05	0.37	1.55	0.25
No. of shareholders	446	240	857	73
Subscribed capital (£)	198,236	335,038	1,233,326	3,200
		t_2		
Share price (% of par)	1.07	0.41	1.95	0.50
No. of shareholders	517	229	888	77
Subscribed capital (£)	183,485	249,233	886,073	3,230

Notes: t_1 and t_2 are from time periods before and after the peak in cycle share prices.
Sources: Share prices obtained from the *Financial Times* and *Investors Monthly Manual*; Number of shareholders and subscribed capital obtained from *Summaries of Capital and Shares.*

RESULTS

Cycle Company Investors

Investors in both samples are first categorized according to occupation.[6] The categories chosen are similar to those of Acheson, Campbell, and Turner (2017), but somewhat more granular. In particular, those working in the cycle industry may have had better information than the average investor; alternatively, they may have experienced some form of familiarity bias. Cycle industry workers are therefore treated as a separate group. In terms of location, investors are categorized according to whether they lived near the London Stock Exchange (i.e., if their address was in the EC or WC postcode region) and whether they lived near the

[6] The replication package for our results can be found at Quinn and Turner (2025).

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Birmingham Stock Exchange (if their address was in the B postcode region). While most cycle share activity took place in Birmingham, price lists reveal that a reasonably large number of trades also took place in London.[7] Proximity to company offices is also used as a variable for cycle firms as part of a later analysis, but the control firms lacked sufficient variation: 9 of the 11 were based in EC/WC, with the others based in the adjacent SW London postcode.

We report both changes in the proportion of shareholders with a particular characteristic and changes in the proportion of capital held by shareholders with that characteristic. Proportion of shareholders provides an indication of the types of people who entered and exited firms at various stages of the bubble. Proportion of capital held is an economically meaningful measure of where the money, in the aggregate, was coming from. However, it is a much noisier measure, as it can be affected by a small number of very large shareholders. As a result, calculating a realistic measure of statistical significance is only feasible for the proportion of shareholders.

The statistical significance of the difference between t_1 and t_2 investor cohorts, for the control and cycle company samples respectively, can be estimated by modeling whether an investor had a particular characteristic or not as a binomial distribution, then using Pearson's chi-squared test to determine whether this probability was the same at t_1 as at t_2. This provides an estimate of the probability that an observed change in the proportion of shareholders with characteristic i occurred due to random variation. Formally, the null hypothesis of:

$$P_{i,\,t=1} = P_{i,t=2} \tag{1}$$

is tested against the two-sided alternative:

$$P_{i,\,t=1} \neq P_{i,t=2} \tag{2}$$

where i represents a particular characteristic (e.g., being a gentleman, living beside a stock exchange), and $P_{i,\,t=1}$ and $P_{i,\,t=2}$ are the proportion of investors with that characteristic in samples t_1 and t_2, respectively. The test statistic is:

$$\chi^2 = \sum_{j=1}^{4} \frac{(O_j - E_j)^2}{E_j} \tag{3}$$

where O_j equates to the number of observations of type j, with each j representing a particular combination of the time of the observation, t,

[7] See, for example, *Financial Times*, "The Cycle Share Market," 30 April 1897.

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and whether the investor has the characteristic *i*. E_j is the proportion of investors in the pooled sample that have the characteristic *i*. This test statistic asymptotically approaches the chi-squared distribution with one degree of freedom and is thus appropriate given the size of our sample (Rao and Scott 1981).

The change in cycle company shareholder clientele can then be compared to the change in control group shareholder clientele using a two-way fixed effects (TWFE) difference-in-differences approach, where the share price crash (which affected only cycle companies) is considered a treatment. To calculate statistical significance, we run a series of logistic regressions in the form of:

$$y_i = \beta_0 + \beta_1 t_2 + \beta_2 C + \beta_3 C t_2 \tag{4}$$

where y_i is the prevalence of characteristic *i*; t_2 is whether shareholder *i* was in the post-crash sample (the time fixed effect); *C* is whether the shareholder held shares in a cycle company (the treatment fixed effect); and Ct_2 is an interaction of the two terms. The estimated effect of the cycle share crash on the prevalence of characteristic *i* is given by the coefficient β_3, and the statistical significance is obtained from its associated robust standard error.

The results of this analysis are shown in Table 3. Pre-crash cycle company investors were, relative to the control group, more likely to be merchants, manufacturers, or engineers, or working in the finance or retail industries. They were less likely to be gentlemen or women. The latter is especially notable since contemporary sources noted that bicycles were particularly popular with women at this time (Rubinstein 1977). These results are consistent with investors being drawn from groups that previous research has associated with speculative or risky investments (Acheson et al. 2021; Acheson, Campbell, and Turner 2017; Maltby and Rutterford 2006; Rutterford and Maltby 2006; Rutterford et al. 2011). Occupational groups that were stereotypically thought of as "naïve," such as farmers, manual laborers, and clergymen, do not appear to have been heavily involved in the bubble. Initial variation in proximity to stock exchanges is largely due to regional differences between industries: relative to the control group, fewer cycle investors were based near the London Stock Exchange, but many more were based near the Birmingham Stock Exchange.

How did the shareholder clientele of cycle companies change during the bubble? Firstly, there was a drop in the proportion of cycle industry workers, suggesting some use of inside knowledge to sell shares in

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TABLE 3
CHANGE IN SHAREHOLDER CLIENTELE FOR CYCLE AND CONTROL COMPANIES

Occupation	Cycle Companies: Percent of Shareholders (*t1*)	Cycle Companies: Percent of Shareholders (*t2*)	Control Companies: Percent of Shareholders (*t1*)	Control Companies: Percent of Shareholders (*t2*)	Difference in Differences
	(1)	(2)	(3)	(4)	(5)
Cycle industry	3.3	2.0**	0.0	0.0	—
Stockbrokers	1.1	1.4	1.3	0.5	1.03**
Banking and finance	5.8	4.7**	3.2	2.1**	0.08
Companies	0.1	0.1	0.5	0.3	0.21
Upper management and secretaries	5.5	3.7**	4.2	2.2**	0.12
Merchants, agents, and commercial	12.6	9.1**	8.4	5.2**	–0.31
Legal workers	4.2	4.0	4.2	3.7	0.35
Public sector	0.8	0.5	1.5	1.3	–0.10
Health services	3.7	3.0*	2.7	2.6	–0.60
News media	0.3	0.3	0.3	0.1*	0.23
Education	2.0	1.5	0.7	0.6	–0.37
Clerks and assistants	5.9	4.5**	4.2	2.2**	0.66**
Police	1.3	1.2	0.1	0.0	–0.10
Retail	5.4	4.5*	2.2	1.5*	–0.29
Gentlemen and nobility	15.6	31.0**	28.4	30.6*	12.99**
Non-cycle manufacturers and engineers	9.1	7.8*	3.6	2.1**	0.15*
Skilled and semi-skilled trades	6.8	5.4	4.6	3.5*	–0.30
Agriculture	0.8	0.7	0.5	0.7	–0.35
Manual laborers and servants	2.1	1.4**	2.5	1.5**	0.35
Military	1.3	1.2	2.2	3.2**	–1.04
Married women	4.8	4.4	10.2	13.3**	–3.57**
Spinsters	5.0	4.8	9.8	14.9**	–5.44**
Widows	1.0	1.2	3.0	4.8**	–1.68
Clergy	1.6	1.4	1.9	2.2	–0.58
Deceased	0.0	0.0	0.1	1.0**	–0.14
Retired	0	0.0	0.0	0.0	0.00
Near London Stock Exchange	3.2	4.5**	15.9	12.0**	5.19**
Near London or Birmingham Stock Exchange	21.7	26.8**	16.7	12.9**	8.91**

* = Significant at the 5 percent level.
** = Significant at the 1 percent level.
Notes: N=12,167 for cycle firms and 8,182 for control firms. The statistical significance reported in Columns (2) and (4) is from Pearson's chi-squared tests, and in Column (5) it is from the two-way difference in differences regression.
Source: *Summaries of Capital and Shares*.

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advance of the crash. There were also statistically significant falls in the proportion of financiers, upper management, merchants, health workers, clerks, retail workers, and manual laborers (though some of these changes are not especially economically significant). The only occupational group for which there was a statistically significant increase in the proportion of shareholders, from 15.6 percent to 31.0 percent, was gentlemen and nobility. There was also a statistically significant increase in the proportion of shareholders living beside the London and Birmingham Stock Exchanges. These results suggest that members of several other occupational groups sold shares to rentiers and those living near stock exchanges, before or during the crash. As Table 1 shows, these sales would often have been for a healthy short-term profit.

Analysis of the control group reveals that, in most cases, falls in the proportion of shareholders from a particular occupational group would likely have happened in the absence of a bubble. The decreases in financiers, upper management, merchants, and retail workers are common across both samples. In the case of clerks and manufacturers, the pattern of a decrease across time was present in both samples, but the decrease in the control group was somewhat larger. However, there is a notable difference in the occupations that replaced these shareholders: whereas cycle company shares were overwhelmingly sold to gentlemen, control company shares were more often sold to women and members of the military. Female investors appear to have entered more successful firms as they became more established, while avoiding speculative bicycle shares both before and after the crash. The scale of changes in the proportion of gentlemen and female investors after establishment is very similar to that observed in the more general sample of Acheson, Campbell, and Turner (2017, p. 619), suggesting that the patterns observed are reasonably representative of the wider market.

Control companies had few Birmingham-based investors, but the proportion of investors based beside the London Stock Exchange decreased over time—in contrast to the cycle firms. This is most likely attributable to a dispersion effect: initial shareholders were disproportionately drawn from central London, where both the company headquarters and stock exchange were based, and the shareholder base then spread out geographically over time. This trend makes the increasing concentration of cycle company shareholders around exchanges all the more striking.

Table 4 shows the proportion of shares held in cycle companies and control companies across the two registers. This measure is somewhat noisy, but it can be seen that the occupational trends identified in Table 3 generally hold when looking at shares held rather than the number of

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TABLE 4
PROPORTION OF SHARES HELD BY OCCUPATION, STATUS AS DIRECTOR OF THE COMPANY, AND PROXIMITY TO STOCK EXCHANGE

	Cycle Companies		Control Companies	
	Percent of Shares t_1	Percent of Shares t_2	Percent of Shares t_1	Percent of Shares t_2
Cycle industry	10.2	6.5	0.0	0.0
Stockbrokers	2.0	1.5	0.6	0.4
Banking and finance	4.0	7.1	2.7	1.4
Companies	9.9	6.9	7.3	4.6
Upper management and secretaries	15.9	8.9	6.7	3.4
Merchants, agents, and commercial	7.3	6.7	10.6	6.0
Legal workers	3.8	4.3	6.3	9.2
Public sector	1.0	0.2	5.7	8.4
Health services	1.6	1.8	4.0	2.6
News media	0.1	0.1	0.1	0.0
Education	0.2	0.2	0.1	0.2
Clerks and assistants	3.1	0.9	0.5	0.6
Police	0.1	0.2	0.0	0.0
Retail	1.2	1.5	0.5	0.7
Gentlemen and nobility	15.9	27.2	40.7	29.6
Non-cycle manufacturers and engineers	12.1	13.4	2.7	1.5
Skilled and semi-skilled trades	5.4	3.8	1.6	1.9
Agriculture	0.3	0.4	0.1	0.2
Manual laborers and servants	0.4	0.2	0.6	0.9
Military	0.9	1.8	4.0	2.0
Married women	2.5	2.9	2.5	19.1
Spinsters	1.0	1.9	1.3	3.4
Widows	0.7	1.1	0.9	1.8
Clergy	0.2	0.3	0.3	0.8
Deceased	0.0	0.0	0.2	1.2
Retired	0.0	0.0	0.0	0.0
Directors	25.6	20.1	16.1	20.6
Near London Stock Exchange	14.7	15.5	32.8	26.4
Near London or Birmingham Stock Exchange	42.1	38.9	34.6	28.2

Source: *Summaries of Capital and Shares*.

shareholders. There is a fall in the proportion of shares held by cycle industry insiders, suggesting that informational advantages were, generally speaking, more salient than familiarity bias when it came to the decision to sell. There is a notable increase in the proportion of shares held by gentlemen and nobility. In the control sample, the proportion of shares

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held by gentlemen falls over time, and there is a large increase in the proportion of shares held by married women.

Analyzing the proportion of shares held also allows us to examine the behavior of company directors, who are identified by name from *Stock Exchange Yearbooks* and cross-referenced with shareholder registers. Directors typically sold shares in their own cycle companies before or during the crash, but increased their holdings of relatively successful control companies. This is consistent with the findings of Braggion and Moore (2013), who find that while directors did not routinely trade on insider information, they did tend to sell prior to periods of particularly poor stock market performance.

In terms of location, the proportion of shares held by those near stock exchanges is much higher than the proportion of shareholders across both samples, suggesting that exchange-adjacent shareholders held substantial numbers of shares. The proportion of cycle shares held by investors based near the London Stock Exchange increased during the bubble, but the proportion held by investors near the Birmingham Stock Exchange decreased (albeit from a very high base). This could be because exchange-adjacent investors entering cycle firms were doing so for relatively small amounts and may even have been purchasing shares from other local investors.

Transfer Data and Investor Activity

The trading behavior of money-losing investors can be examined further through transfer data. As Griffin et al. (2011) note, a given group of investors can lose money during a bubble in two ways. Firstly, they can supply fewer shares when prices are at their peak; that is, they are less likely to sell at the right time. Secondly, they can demand more shares when prices are high; that is, they are more likely to buy at the worst possible time. Transfer data allows us to track the level of selling activity, and thereby investigate the extent to which occupational and geographic groups were supplying shares to the market during the crash. The purchasers of shares were not recorded, so demand is not directly observable; however, it can be deduced through a combination of recorded sales and clientele changes between t_1 and t_2. For example, if a given occupational group increased their holdings despite accounting for a large number of sales, the increase in their number must be demand-driven.

Since transfer records were only reported by a minority of cycle companies, we first restricted the sample to the 10 cycle firms in our main sample. This results in a sample of 9,134 shareholders, 7,708 of whom

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provided an occupation. The patterns of ownership within this sub-sample are broadly similar to those observed in the full sample. The transfer data of these companies consists of 1,996 transfers conducted by 1,568 unique individuals. For these companies, we have a record of ownership pre-crash, a record of sales, and a record of ownership post-crash. We can therefore repeat the analysis of shareholder identities, testing for whether certain groups were over-represented in the sellers group relative to initial shareholders. All of the control companies recorded transfer data, so these companies are again used to provide a baseline for investor behavior.

The results of this analysis for the cycle companies are shown in Table 5. It can be seen, firstly, that the t_1 and t_2 clientele for this sub-sample are similar to those for the full sample, reported in Table 3. The transfer data reveals that the group that conducted the most sales was gentlemen and nobility—also the group that increased its ownership of cycle shares by the most during the bubble. This indicates that this group performed poorly in aggregate because it was extremely active in the secondary market at a time when cycle share prices were close to their peak. Although they were more likely to sell than any other group, they were also much more likely to buy, suggesting that this group engaged in substantial speculative investment.

Location data for the cycle companies contains enough variation to decompose the effects of proximity to a stock exchange and proximity to a company office. It can be seen that, like gentlemen, investors based near a stock exchange increased their holdings despite being very active sellers. Investors based near a company office were more likely to reduce their holdings, particularly if they were not also near a stock exchange (although the effect size is small). This suggests that investors near a company office may have used privileged information to exit prior to the crash. However, investors beside a company office but not a stock exchange are not over-represented in the sample of sellers, indicating that this informational advantage manifested itself in being less likely to buy on the eve of the crash.

The final rows examine gentlemen who lived near a stock exchange in order to identify any interaction effects. This group accounted for only 1.86 percent of shareholders at time t_1. However, it accounted for 19.06 percent of transfers and 8.93 percent of t_2 shareholders, suggesting a positive interaction effect. The combination of being a gentleman and living near a stock exchange made it much more likely that an investor would become an active trader and that they would end up holding cycle shares after the crash. While this finding is statistically significant for both the London and Birmingham Stock Exchanges, the effect of living near the Birmingham Stock Exchange was considerably larger.

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TABLE 5
OCCUPATIONS OF PRE-PEAK SHAREHOLDERS, SELLERS, AND POST-PEAK SHAREHOLDERS AS A PERCENTAGE OF ALL SHAREHOLDERS (CYCLE COMPANIES WITH TRANSFER DATA)

Occupation/Location	Percent of Shareholders t_1	Percent of Sellers	Percent of Shareholders t_2
Cycle industry	3.14	2.30	1.60**
Stockbrokers	0.98	4.29**	1.20
Banking and finance	5.34	4.69	4.48
Companies	0.03	0.24	0.04
Upper management and secretaries	5.38	4.29	3.39**
Merchants, agents, and commercial	10.89	8.82*	8.02**
Legal workers	3.62	4.05	3.28
Public sector	0.91	0.56	0.61
Health services	3.58	1.75**	2.67*
News media	0.24	0.00	0.23
Education	2.30	0.64**	1.58*
Clerks and assistants	6.29	3.34**	4.29**
Police	0.85	0.00**	0.82
Retail	5.24	2.86**	4.57
Gentlemen and nobility	14.50	38.68**	33.85**
Non-cycle manufacturers and engineers	8.22	8.50	6.69*
Skilled and semi-skilled trades	8.08	4.69**	5.54**
Agriculture	0.85	0.32*	0.72
Manual laborers and servants	2.81	1.03**	1.75**
Military	1.45	0.40**	1.41
Married women	5.88	3.81**	4.99
Spinsters	6.22	2.54**	5.68
Widows	1.12	0.87	0.97
Clergy	2.10	1.35	1.60
Beside Birmingham/London Stock Exchange	19.70	51.66**	24.09**
Beside Company Office	25.33	45.03**	22.55**
Beside Exchange AND Beside Company Office	12.31	32.59**	13.18
Beside Exchange but NOT Beside Company Office	7.38	19.07**	10.91**
Beside Company Office but NOT Beside Stock Exchange	13.02	12.44	9.37**
Beside neither Exchange nor Company Office	67.29	35.91**	66.54
Unknown	0.31	0.19	0.91**
Gentlemen/nobility AND Beside London OR Birmingham Stock Exchange	1.86	19.06**	8.93**
Gentlemen/nobility AND Beside London Stock Exchange	0.95	8.10**	2.55**
Gentlemen/nobility AND Beside Birmingham Stock Exchange	0.91	10.96**	6.38**

* = Significant at the 5 percent level.
** = Significant at the 1 percent level.
Notes: Results are for the subset of cycle firms that recorded transfers. The statistical significance reported is for Pearson chi-squared tests for difference from the t_1 sample.
Source: National Archives, *Summaries of Capital and Shares*.

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TABLE 6
LOGIT REGRESSION ON THE CHARACTERISTICS
OF POST-CRASH CYCLE SHAREHOLDERS

	(1)	(2)
Located near an Exchange	0.181**	0.036
	(0.053)	(0.057)
Located near the company office	–0.117*	–0.107*
	(0.054)	(0.054)
Gentleman	0.882**	0.713**
	(0.046)	(0.052)
Gentleman*Exchange		0.707**
		(0.112)
Urban	0.015	0.042
	(0.044)	(–0.044)
Constant	0.102**	0.349**
	(0.032)	(–0.031)

* = Significant at the 5 percent level.
** = Significant at the 1 percent level.
Notes: Robust standard errors in parentheses.
Source: National Archives, *Summaries of Capital and Shares*.

This possibility can be examined formally using a logit regression, in which the dependent variable is whether a given investor in the full sample held shares at time t_2 rather than at time t_1. Our dependent variables are whether an investor was located beside an exchange, whether they were located beside the company office, whether they were a gentleman, and an interaction term for gentlemen who also lived beside an exchange. Whether an investor lived in an urban area is also included as a control variable. The results are shown in Table 6. Notably, when the interaction term is included, the significance of living near a stock exchange disappears. In other words, living near a stock exchange was only a disadvantage for gentlemen; for other demographics, it had no impact on the probability of losing money during the bubble.

The results of the equivalent analysis for the control group are shown in Table 7. Gentlemen who lived near a stock exchange are also significantly over-represented among sellers in this group, suggesting that this was a demographic that was very active in share markets generally rather than being specifically drawn into investing in cycle shares. However, in contrast to the cycle firms, the proportion of gentlemen living near the London Stock Exchange in these firms falls over time. While they were highly active in both markets, they increased their holdings in companies

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TABLE 7
CHARACTERISTICS OF PRE-PEAK SHAREHOLDERS, SELLERS, AND POST-PEAK SHAREHOLDERS AS A PERCENTAGE OF ALL SHAREHOLDERS (CONTROL COMPANIES)

Occupation/Location	Percent of Shareholders t_1	Percent of Sellers	Percent of Shareholders t_2
Cycle industry	0.0	0.0	0.0
Stockbrokers	1.3	1.8	0.5
Banking and finance	3.2	0.9*	2.1**
Companies	0.5	1.5*	0.3
Upper management and secretaries	4.2	1.8	2.2**
Merchants, agents, and commercial	8.4	5.5*	5.2**
Legal workers	4.2	3.4	3.7
Public sector	1.5	1.8	1.3
Health services	2.7	1.5	2.6
News media	0.3	0.0	0.1*
Education	0.7	0.0	0.6
Clerks and assistants	4.2	2.7	2.2**
Police	0.1	0.0	0.0
Retail	2.2	0.9	1.5*
Gentlemen and nobility	28.4	43.3**	30.6*
Non-cycle manufacturers and engineers	3.6	0.9*	2.1**
Skilled and semi-skilled trades	4.6	1.5*	3.5*
Agriculture	0.5	0.3	0.7
Manual laborers and servants	2.5	1.5	1.5**
Military	2.2	1.5	3.2**
Married women	10.2	7.9	13.3**
Spinsters	9.8	6.7	14.9**
Widows	3.0	2.7	4.8**
Clergy	1.9	0.6	2.2
Deceased	0.1	11.0**	1.0**
Retired	0.0	0.0	0.0
Beside London Stock Exchange	15.9	29.6**	12.0**
Gentlemen/nobility AND Beside London Stock Exchange	4.08	14.3**	3.24*

* = Significant at the 5 percent level.
** = Significant at the 1 percent level.
Notes: The statistical significance reported is for Pearson chi-squared tests for difference from the t_1 sample.
Source: National Archives, *Summaries of Capital and Shares*.

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that went on to fail and decreased their holdings in companies that went on to become well-established.

Discussion

The groups of investors who were more likely to exit or reduce their holdings in cycle firms prior to or during the crash were those working in the industry, either as cycle agents, cycle manufacturers, or directors of cycle companies. In terms of location, the investors who reduced their holdings were those living beside a company office but not beside a stock exchange. These results suggest that systematically profiting from a bubble is possible, but often requires privileged information or proximity to the epicenter of the industry involved. This is consistent with theories of bubbles in which mispricing is not immediately corrected by arbitrageurs because informational advantages present larger profit opportunities (Abreu and Brunnermeier 2002).

The occupational group that performed worst was gentlemen, a group defined by the fact that they had enough property that they did not need to work. This meant that they had sufficient time and money to trade speculative stocks, and transfer data shows that they were particularly active buyers and sellers. Gentlemen were much more likely to be left holding worthless shares if they lived near a stock exchange, perhaps because this made frequently trading cycle shares much more convenient.

While narratives of bubbles often emphasize the role of naïve investors, the investors most at risk during the cycle mania were not necessarily the least experienced or informed. Our results are more consistent with theories in which the investors that lose money are those most prone to behavioral biases, particularly overconfidence and familiarity. Døskeland and Hvide (2011) argue that the underperformance associated with these biases can be explained by investors falsely believing themselves to have superior information. In this instance, proximity to a stock exchange may have provided gentlemen with enough information to attempt to speculate in cycle shares, but not enough information to do so successfully.

The control group results reveal that gentlemen living near a stock exchange also reduced their holdings of stocks that went on to be relatively successful, suggesting a level of poor trading performance that extended beyond the bubble in bicycle shares. They were also disproportionately active traders in both sets of firms. Women, who previous research has shown are much less overconfident traders, accounted for a very small proportion of bubble investors, but significantly increased their holdings of control companies; they also traded less frequently than

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men. Our results are therefore consistent with Barber and Odean (2001), who argue that men experience lower returns than women due to an overconfidence-induced propensity to trade more often.

CONCLUSION

This paper uses a unique dataset on holders of British cycle shares during the asset price reversal of 1896–1900 to examine investor performance during a bubble. Three main conclusions are reached. Firstly, consistent with the work of Campbell and Turner (2012) and Carlos and Neal (2006), the bubble was not driven by extremely inexperienced or naïve investors. Instead, investors in cycle shares tended to come disproportionately from groups associated with a preference for relatively risky investments.

Secondly, investors who sold shares prior to the crash, many of whom would have done so for a profit, tended to come from groups that might be expected to have privileged information. Company directors, cycle industry insiders, and those based near the company's office all systematically reduced their holdings during the crash. This supports the findings of Braggion and Moore (2013), Sentis (2009), and Shen, Hui, and Fan (2021), who find evidence of insiders exiting firms prior to a crash in modern and historical stock and housing markets.

Thirdly, we provide new insights into which investors lose money during a bubble. The investors left holding cycle shares that were almost worthless after 1897 were predominantly gentlemen based near a stock exchange who were active on secondary markets. These were unlikely to have been the least informed investors; rather, they were investors with a moderate level of information, which left them susceptible to overconfidence. They were also investors with sufficient time and money to buy and sell shares frequently, and proximity to a stock exchange would have made this particularly convenient.

Our results are particularly important in light of the recent demand-system asset pricing literature, which argues that securities prices are responsive to changes in investors' demand, even in the absence of relevant news (Cortes, Cunha, and Barbosa 2022; Gabaix and Koijen 2021; Koijen and Yogo 2019, 2020). This implies that changes in the level of investment from particular demographic groups can be an independent causal factor in asset price bubbles and crashes. Robust identification of this effect is beyond the scope of this paper, but the insight into how shareholder clientele changed during the bicycle mania may provide a platform for future theoretical and empirical work into the role of demand-side factors in bubbles more generally.

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Appendix

APPENDIX TABLE 1
CYCLE COMPANY SHAREHOLDER RECORDS

Company	Shareholder Register t_1	Shareholder Register t_2	Total Shareholders (t_1+ t_2)	Total Shares (t_1+ t_2)	Reason for Winding Up
Accles	16/10/1896	31/12/1898	248	294,642	Bankruptcy
Appleby Joseph	11/08/1896	05/07/1897	157	129,424	Bankruptcy
Austral Agency	08/03/1897	08/03/1898	272	220,000	Bankruptcy
Belle Vale Tube	04/01/1897	21/04/1898	202	39,845	Bankruptcy
Boudard Peveril	19/11/1894	25/12/1898	622	8,791	Unknown - disbanded in 1901
Bown	31/01/1897	24/02/1899	293	101,040	Bankruptcy
Brookes Cycle	18/09/1896	31/12/1898	466	120,000	Bankruptcy
Claremont	30/10/1896	18/01/1898	894	150,145	Voluntary with share price at 3% of par value
Concentric Tube	03/09/1896	25/06/1897	596	149,000	Bankruptcy
Coventry Cross	22/09/1896	27/12/1898	442	72,705	Reconstructed with share price at 18% of par value
Dunlop J B Fittings	12/04/1897	12/10/1900	222	57,029	Unknown – wound up in 1902 with share price at 13% of par value
Empire	21/12/1896	22/11/1898	312	49,667	Bankruptcy
Endurance Tubes	26/09/1896	10/12/1897	502	124,902	Bankruptcy
Humber and Goddard	07/10/1896	01/02/1899	278	144,638	Voluntary with share price at 25% of par value
Larue	28/10/1896	02/02/1898	371	145,795	Unknown – wound up unlisted in 1900
Mutual	27/11/1895	14/01/1898	1,742	250,384	Voluntary with share price at 50% of par value
New Seddon	17/09/1896	30/04/1897	3,358	498,806	Reconstructed in 1899 with share price at 5% of par value
Puncture Proof	11/05/1896	06/02/1899	805	59,294	Merged in 1898 with share price at 18% of par value
Quadrant	02/03/1896	24/10/1900	248	97,619	Wound up in 1908 with share price at 25% of par value
Raleigh	05/06/1896	14/12/1897	2,042	400,035	Reconstructed in 1899 with share price at 5% of par value
Reliance	10/09/1896	08/04/1898	266	48,479	Bankruptcy
Rosser Brake	20/02/1897	14/01/1898	42	58,014	Unknown – wound up in 1898
Sanspareil	06/04/1897	09/02/1899	215	40,000	Voluntarily wound up in 1898
Simpson's Chain	31/12/1895	31/12/1897	1,393	471,012	Bankruptcy
Truffault	13/09/1896	13/11/1897	1,000	158,379	Reconstructed in 1898 with share price at 18% of par value
Total			16,988	3,889,645	

Source: *Summaries of Capital and Shares.*

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APPENDIX TABLE 2
CONTROL COMPANY SHAREHOLDER RECORDS

Company	Shareholder Register t_1	Shareholder Register t_2	Total Shareholders (t_1+ t_2)	Total Shares (t_1+ t_2)
Anglo Sicilian Sulphur	20/11/1896	25/10/1901	794	2,119,399
Aux Classes Laborieuses	26/04/1898	26/07/1900	1,522	148,099
De Keysers Royal Hotel	18/05/1897	14/02/1902	816	46,914
Dooars	04/02/1895	11/05/1896	150	6,430
Hood and Moores Stores	10/04/1896	05/07/1901	827	290,241
Burlington Hotels	14/04/1897	04/02/1902	793	270,150
Davies MC Karri and Jarrah	03/03/1898	03/12/1903	1,745	663,635
Elysee Palace Hotel	26/05/1897	25/12/1902	528	339,689
Welfords Surrey Dairies	27/04/1897	24/03/1902	1,526	149,198
Bournemouth and Poole Electric Supply	16/04/1898	08/04/1903	784	34,360
South London Electric	04/04/1898	01/04/1903	1,101	130,812
Total			10,586	4,198,927

Source: *Summaries of Capital and Shares.*

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APPENDIX TABLE 3
COMPANY SHARE PRICES AT t_1 AND t_2

Company	Share Price t_1	Share Price t_2	Return
Accles	0.75	0.05	–0.93
Appleby Joseph	1.23	1.06	–0.13
Austral Agency	1.50	0.05*	–0.97
Belle Vale Tube	1.03	0.05*	–0.95
Boudard Peveril	7.00	0.69	–0.90
Bown	1.05	0.40	–0.62
Brookes Cycle	1.00	0.13	–0.88
Claremont	1.13	0.30	–0.73
Concentric Tube	1.20	0.76	–0.36
Coventry Cross	1.15	0.39	–0.66
Dunlop J B Fittings	1.10	0.13	–0.89
Empire	1.01	0.09	–0.91
Endurance Tubes	1.06	0.26	–0.75
Humber and Goddard	0.65	0.25	–0.62
Larue	0.45	0.25*	–0.44
Mutual	0.25	0.05	–0.80
New Seddon	0.44	0.30	–0.31
Puncture Proof	1.30	0.18	–0.86
Quadrant	1.03	0.45	–0.56
Raleigh	1.50	0.80	–0.47
Reliance	1.26	0.13	–0.90
Rosser Brake	1.04	0.25*	–0.76
Sanspareil	1.09	0.25*	–0.77
Simpson's Chain	1.03	0.19	–0.82
Truffault	0.55	0.13	–0.77
Anglo Sicilian Sulphur	1	0.68	–0.32
Aux Classes Laborieuses	0.95	0.85	–0.11
De Keysers Royal Hotel	1.2	1	–0.17
Dooars	1.55	1.95	0.26
Hood and Moores Stores	0.87	0.5	–0.43
Burlington Hotels	1.5	1.5	0.00
Davies MC Karri and Jarrah	1.12	0.75	–0.33
Elysee Palace Hotel	0.25	1	3.00
Welfords Surrey Dairies	1.5	1.37	–0.09
Bournemouth and Poole Electric Supply	1	1.4	0.40
South London Electric	0.662	0.8	0.21

Notes: All prices expressed as a percentage of par value. *Indicates estimates based on last reported share prices and winding up records.
Sources: *Birmingham Daily Mail, Financial Times,* and *Investor's Monthly Manual.*

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